

Manuscript Terminology Consistency Plan

2026-06-17 19:37 PDT

A plan to align the terminology in all `report.Rmd` and `report-slim.Rmd` manuscripts with the project lexicon (`docs/29-nof1-precision-medicine-lexicon.md`) and with each other.

Contents

Execution status (2026-07-30)	3
Scope and method	3
1. Decisions to lock before editing	3
2. Tier 1 - project-wide standardisations (safe, high value)	5
3. Tier 2 - per-paper consistency fixes	6
4. Tier 3 - glossing / definition additions	9
5. Cross-file consistency (full vs slim)	10
6. Out of scope: substantive issues to flag to the author	10
7. Execution workflow	10
8. Execution record (2026-07-30)	11

List of Tables

1	Terminology decisions with recommended defaults and supporting corpus evidence	4
2	Per-paper terminology consistency fixes with notes	6

Execution status (2026-07-30)

Executed with D1 locked to US English. Tier 1 was applied to all 21 `report*.Rmd` files (the plan was written against 13; six variants have been added since, and paper 11 did not exist). Tier 2, Tier 3, and the Section 5 full-versus-slim pass were applied where the targets still existed; most had already converged through intervening revisions. Section 6 items remain open and are for the author.

What changed, and what was already clean, is recorded at the end of this file. All 21 manuscripts were re-rendered and build.

Scope and method

Thirteen manuscript files across ten papers were reviewed against the lexicon (prose, headings, captions, and abstracts only; R code chunks and in-model variable names such as `c.bm`, `Dbc`, `carryover_t1half` were excluded). Papers 01, 02, and 06 have both a full and a slim variant; the variants were reviewed together for cross-file consistency. The `rgt` paragraph blocks are still Lorem ipsum placeholders and were not reviewable; a second pass will be required once the author writes that prose.

Finding in one line. No statistical term is *wrong*; the issues are (i) inconsistent surface forms for the same concept across and within papers, (ii) acronyms used before first-use expansion, and (iii) a few colloquial coinages and one genuinely narrow lexicon term (“washout”) used loosely. Three *substantive* (non-terminological) issues surfaced incidentally and are listed separately in Section 6; they require author judgement and must not be folded into a mechanical edit.

Important execution constraint. These files live under `~/Library/CloudStorage/` (Dropbox). Per the project guardrails, in-place stream editors (`sed -i`, `perl -i`, etc.) must NOT be used on cloud-mounted paths. All edits must go through the Edit tool (which is path-aware) or the copy-to-`/tmp`, edit, mv-back pattern. Global substitutions should be applied per file with `replace_all`, never with a recursive `sed -i`.

1. Decisions to lock before editing

These choices propagate everywhere, so confirm them first. Recommended defaults are evidence-based (current corpus counts in parentheses).

Table 1: Terminology decisions with recommended defaults and supporting corpus evidence

#	Decision	Recommended default	Evidence
D1	Spelling convention	US / -ize, -or, modeling (locked 2026-07-30; overrides the frequency-based recommendation below)	The corpus leans British (modelling 73 vs modeling 43; characterise 26; behaviour 18 vs 3; generalised 16 vs 8), but frequency records drift, not a decision. The project-wide standard is US English, so the direction of travel is British -> US.
D2	Casing of “Type I error”	Type I error (cap T, cap I, no hyphen)	Type I error 121 vs type I 61 vs type-I 45
D3	Biomarker-interaction surface form	biomarker-treatment interaction	biomarker-treatment vs biomarker-by-treatment 45; matches lexicon §10 headword
D4	Monte Carlo SE form	spell Monte Carlo standard error (MCSE) at first use, then MCSE	MCSE 281, MC SE 59, spelled 57, “Monte Carlo SE” 25
D5	“washout” usage	reserve washout for an inserted treatment-free clearance gap; use off-drug period / occasions for the general post-discontinuation span	off-drug 279 vs washout 92; the papers’ own thesis is that carryover contaminates these spans, so “washout” is self-contradictory there (lexicon §2)

#	Decision	Recommended default	Evidence
D6	Precision vs personalised medicine	precision medicine (primary), “personalised medicine” only as a glossed synonym	precision 11 vs personalised 4; lexicon §10 headword
D7	Participant unit noun	participant in prose; use “subject”/“cluster” only where theory requires (e.g. sandwich estimator)	papers 05/08 use participant; 10 mixes cluster/subject/participant

2. Tier 1 - project-wide standardisations (safe, high value)

Apply across all 13 files. Each is a low-risk, presentational fix that does not alter meaning. Apply with the Edit tool (**replace_all** per file), checking each hit in context (especially D5, which is contextual, not a blind replace).

1. **Type I / Type II error casing** (D2). Normalise **type I error**, **type-I error**, **type-I**, **Type-I** -> **Type I error**; likewise **Type II**. Affects 02, 04, 05, 07, 08, 09, 10. (~106 non-canonical instances.)
2. **Biomarker-treatment interaction** (D3). **biomarker-by-treatment interaction** -> **biomarker-treatment interaction**; also normalise **treatment-by-biomarker**, **biomarker:treatment**, “predictive interaction” where they denote the same estimand. Affects 02, 03, 06, 08, 09, 10. (~45 instances.)
3. **MCSE** (D4). **MC SE** and **Monte Carlo SE** -> **MCSE**; ensure one spelled-out first use per file. Affects 01, 02, 07 (and any ADEMP paper). (~84 instances.)
4. **HTE first-use + label**. Spell **heterogeneity of treatment effects** (HTE) at first use in every paper that uses the concept, then HTE. Replace loose paraphrases (“treatment-response heterogeneity”, “individual treatment heterogeneity”, “between-person response variability”) with HTE or with “patient-by-treatment interaction” where the variance-component meaning is intended. Critical in 06 (concept central but acronym never introduced) and 01 (used before expansion). Also 03, 04.
5. **“washout” -> “off-drug period/occasions”** (D5). Contextual: change only where the text means the post-discontinuation span in which

carryover decays; keep “washout” where a clearance gap is genuinely inserted. Affects 01, 02, 04. (~92 “washout” hits to triage.)

6. **Acronym first-use expansion.** Ensure each is spelled out at first reader-facing use, then abbreviated: DGP (data-generating process/mechanism), MVN (multivariate normal), LME (linear mixed-effects model), MCAR/MAR/MNAR, OL/BDC/OL+BDC, CO (crossover), PK/PD, BLUP, SEM, IL-6, RM-ANOVA, OFAT. The three-part `bullets/orig` structure aggravates this because a `bullets` block often leads with a bare acronym that the later `orig` block expands; fix by expanding in whichever block appears first.
7. **Spelling normalization** (D1, direction reversed on 2026-07-30). `generalised` -> `generalized`, `modelling` -> `modeling`, `behaviour` -> `behavior`, `parameterise` -> `parameterize`, `characterise` -> `characterize`, `normalise` -> `normalize`, `whilst` -> `while`, `artefact` -> `artifact`, `grey` -> `gray`, etc. Preserve British spelling inside verbatim quotations and in published citation or reference titles. Do not touch the US-English false friends (`exercise`, `revise`, `raise`, `advise`, `promise`, `compromise`, `analyses`, `characteristic`, `programmer`).
8. **Code tokens leaking into prose.** Replace reader-facing code identifiers with their prose terms: `OLBDC` -> `OL+BDC` (02), `modGompertz` -> “modified Gompertz function” in narrative (07; keep `modGompertz` only when referring to the R function). Keep all genuine model-symbol references (`$D_{bc}$`, `$c.bm$`, `bm:Dbc`) as code.

3. Tier 2 - per-paper consistency fixes

Table 2: Per-paper terminology consistency fixes with notes

Paper	Fix	Notes
01	Standardise the drug-indicator naming (“treatment indicator” / “drug indicator” / “drug exposure variable” -> one prose term for the DGP indicator and one for the analysis variable).	Within-paper inconsistency for <code>\$D_{it}\$</code> / <code>Dbc</code> .

Paper	Fix	Notes
02	Settle on ONE label for analysis spec A2 (“matched-decay” / “exposure-weighted” / “continuous- D_{bc} ”); reserve “matched-decay” for the matched DGP cell. Fix “censored” where it means dropped/missing (lexicon §8 vs §11). Standardise “Hybrid N-of-1 design” and “crossover (CO)” (drop “classical/traditional” alternation).	A2 label multiplicity (~25) is the biggest clarity win.
03	Disambiguate unobserved-group vocabulary: use latent class for unobserved groups; reserve subgroup/stratum for observed splits (lexicon §10). Gloss subadditivity and frailty at first use; change “diagnostic biomarker” -> predictive biomarker where differential benefit is meant.	“frailty” is an unglossed third synonym for random effect.

Paper	Fix	Notes
06	Replace colloquial lumped -> one-component / single-indicator (both already used). Define response expectancy once, then use “expectancy” rather than rotating belief/suggestion/suggestibility/optimism. Gloss balanced-placebo (open-hidden) design at first use. Standardise “between-participant”.	See also the Study B/C cross-file bug in Section 6.
07	Capitalise Gompertz consistently in prose (lowercase only as a factor level/code). Remove modGompertz from narrative (Tier 1.8). Add a one-line gloss tying Hill / sigmoid-Emax / indirect-response to the lexicon “Emax / dose-response model” cluster.	Mostly cosmetic but pervasive.
08	Disambiguate main effect: use average treatment effect (ATE) for the population mean effect; keep “main effect” only for the ANOVA-factor sense. Introduce anti-conservative (inflated Type I error) once, then use one term.	“main effect” overload is a genuine ambiguity.

Paper	Fix	Notes
09	Consolidate informative dropout (canonical) over “biased dropout”/“attrition” (keep “biased” only for cell labels). Use ignorable (not “non-informative”) for the MAR component to avoid collision with “noninformative prior”. Replace colloquial estimand drift -> “bias in the estimand” and design surface area -> a defined phrase. Normalise MNAR spelling.	See MCAR_25 flag in Section 6.
10	Choose one orthography for mis-specified/misspecified . Introduce cluster-robust (sandwich) standard error once, then one term. Keep working correlation structure for the GEE object only; use working covariance for the LME structure. Fix participant noun (D7).	Most careful paper; fixes are orthographic.

4. Tier 3 - glossing / definition additions

Add a one-line first-use gloss (no statistical change) for load-bearing terms absent from the lexicon: **subadditivity** (03, 01), **frailty** (03), **balanced-placebo / open-hidden design** (06), **vulnerability window** and **design surface area** (09), **lasagna plot** (02). Consider adding **subadditivity**, **balanced-placebo design**, and **gating function** as new entries to docs/29-...-lexicon.md so future papers inherit a fixed definition.

5. Cross-file consistency (full vs slim)

For papers 01, 02, 06, run a final diff pass so the slim variant uses the same canonical forms and acronym expansions as the full report (e.g. IL-6 is defined in 01 full but not slim; “RM-ANOVA” vs “repeated-measures analogue” differs across 02 full/slim).

6. Out of scope: substantive issues to flag to the author

These are NOT terminology and must not be auto-edited; each needs author judgement because a “fix” could change meaning or a stated result.

1. **Paper 02, report.Rmd Conclusion #1.** States A2 is “the recommended default ... across the full factorial”, which conflicts with the paper’s own data (A2 leads in ~19% of cells) and with the slim variant’s correctly hedged conclusion. (Consistent with the prior project note that the A2-dominance headline is not universal.)
2. **Paper 06, Study B/C naming divergence.** The full report labels the belief-decoupling study “Study B” (sec 7.5) and the coupling pilot the “contaminated-biomarker pilot” (7.4); the slim report calls them “Study C” and “Study B pilot”. Any reader cross-referencing the two files is misdirected. Decide one scheme and apply to both.
3. **Paper 09, MCAR_25 cell label.** Its flat dropout component scales with elapsed time (a design variable), which is MAR, not MCAR, under the lexicon definitions. Renaming would change a stated result label, so confirm the mechanism before any change.

7. Execution workflow

1. Lock decisions D1-D7 (Section 1).
2. Apply Tier 1 globally, one file at a time, via the Edit tool with `replace_all`; triage D5 (“washout”) hit-by-hit. Do not use `sed -i` on these cloud paths.
3. Apply Tier 2 per paper; Tier 3 glosses; Section 5 full-vs-slim diff.
4. Re-render each changed manuscript with the appropriate R environment (note: paper 04 requires the Framework R / kableExtra environment per the project render notes) and confirm it still builds and that the PDF render-timestamp footer is updated.
5. Address Section 6 items separately with the author.

Effort estimate. Tier 1 is mechanical and dominated by D2/D3/D4 (~235 substitutions) plus contextual D5 triage (~92 hits). Tier 2 is a few dozen targeted edits per paper. The re-render and full-vs-slim diff are the main time cost. None of the work changes a number, estimand, or model.

8. Execution record (2026-07-30)

Tier 1 applied across 21 files by a chunk-aware transform (prose rules skip fenced R chunks so that ggplot's `colour=` and dplyr's `summarise()` are untouched; display-string rules such as MC SE are applied inside chunks as well, because those strings are table headers):

- D2 Type I/II casing, D3 biomarker-by-treatment -> biomarker-treatment, D4 MC SE/MC SEs -> MCSE/MCSEs, Tier 1.8 OLBDC -> OL+BDC.
- D1 in the US direction: `modelling`, `generalis*`, `behaviour`, `characteris*` (guarding `characteristic`), `normalis*`, `summaris*` (prose only), `artefact`, `randomis*`, `standardis*`, `dichotomis*`, `ageing`, `sceptic*`, `favour*`, `enrolment`, `signalling`, `programme`, `labelled`, `modelled`, `analyse`, `centre*`, `colour` (prose only), `whilst`, `amongst`.
- 16 of 21 files changed; residual British-looking tokens are all R identifiers (`dplyr::summarise`) and are correct as they stand.

Tier 1 items already satisfied: HTE is expanded at first use in 01 and 06; the acronym sweep (DGP, MVN, LME, MCAR/MAR/MNAR, OL/BDC, CO, PK/PD, BLUP, SEM, RM-ANOVA, OFAT, GEE, ICC) found only five true misses, fixed here: MCAR and MAR in 02, BLUP in 04, DGP and MVN in 11, plus RM-ANOVA in 01-slim and MCAR/MAR in 02-slim for Section 5 parity. ADEMP is treated as a named, always-cited framework and is not expanded.

D5 (washout) triage. 35 hits reviewed individually. All but three are genuine inserted clearance gaps and were kept. The three changed are in paper 02's toy example, where the span is the carryover-contaminated post-discontinuation window: `three-week washout` -> `three-week off-drug period`, the second washout week -> the second off-drug week, throughout the washout -> throughout the off-drug period.

Tier 2, applied: 02, one residual `matched-decay` (M2) reference to the specification changed to `exposure-weighted` (M2) (the `matched-decay cell` retains the term, as the plan specifies).

Tier 2, already converged: 06 (lumped survives only as the math superscript $\beta_{bm}^{\text{lumped}}$, not as prose); 07 (`modGompertz` absent from narrative; lowercase `gompertz` occurs only in citation keys, file paths, and equation labels); 08 (`main effect` absent; `anti-conservative` already single-form); 09 (`attrition`, `estimand drift`, `design surface area`, `non-informative` all absent; `biased dropout` survives only as cell-label description, which the plan permits); 10 (misspecified already uniform, 13/13).

Tier 3 glosses added: `subadditivity` (03), `frailty` (03), `balanced-placebo design` (06). `lasagna plot` has no occurrences.

Not done: the plan's suggestion to add `subadditivity`, `balanced-placebo design`, and `gating function` to `docs/29-...-lexicon.md`; Section 6 sub-

stantive items; and the second terminology pass over the `rgt` blocks, which remain placeholders.

Interaction with the notation pass. This execution followed the symbol-level pass recorded in `analysis/report/NOTATION.md`. The analysis specifications are now M1/M2/M3 in all reader-facing text, which is why this file refers to M2 where the 2026-06-17 draft said A2.

Rendered on 2026-06-17 at 19:37 PDT; execution record appended 2026-07-30. Source: `~/prj/res/36-pmsimstats-ng/pmsimstats-ng/docs/30-terminology-consistency-plan.md`